WORKED SOLUTION

STEP III 1996, Question 2

Pure • Simultaneous Linear Equations

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/96-s3-q2

The key idea

Three linear equations in three unknowns have exactly one solution unless the determinant of the coefficients vanishes. So the first job is to find the values of a that make the determinant zero; for every other a we simply solve. The real content of the question is in the exceptional values, where we must decide, depending on b, whether the equations are inconsistent or have infinitely many solutions.

Solution

Label the equations

$$x + y + az = 2, \tag{1}$$
$$x + ay + z = 2, \tag{2}$$
$$2x + y + z = 2b. \tag{3}$$

When is there a unique solution?

The coefficient matrix has determinant

$$\begin{vmatrix} 1 & 1 & a \\ 1 & a & 1 \\ 2 & 1 & 1 \end{vmatrix} = 1(a-1) - 1(1-2) + a(1-2a)$$
$$= 2a - 2a^2 = 2a(1-a).$$

So there is exactly one solution, whatever b is, when $a \neq 0$ and $a \neq 1$. The cases $a = 0$ and $a = 1$ need separate treatment.

The general case: $a \neq 0, 1$

Subtracting (2) from (1) gives

$$(1-a)y - (1-a)z = 0,$$

and since $a \neq 1$ we may divide by $1-a$ to get $y = z$. Putting $z = y$ into (1) and (3):

$$x + (1+a)y = 2, \qquad 2x + 2y = 2b, \quad \text{i.e.} \quad x + y = b.$$

Subtracting the second from the first gives $ay = 2 - b$, and since $a \neq 0$,

$$y = z = \frac{2-b}{a}, \qquad x = b - \frac{2-b}{a} = \frac{(a+1)b - 2}{a}.$$

Substituting back into (1), (2) and (3) confirms that this is a solution, and by the argument above it is the only one.

The case $a = 0$

The equations become

$$x + y = 2, \qquad x + z = 2, \qquad 2x + y + z = 2b.$$

Adding the first two gives $2x + y + z = 4$. So the left-hand side of the third equation must equal both 4 and $2b$.

> **Ask yourself** Does a zero determinant mean "no solution"?

Not by itself: it means the equations are dependent, so either they contradict each other or one of them adds no new information. Which happens depends on the right-hand sides, here on b.

- If $b \neq 2$, then $4 = 2b$ is false, so there is **no solution**.
- If $b = 2$, the third equation is just the sum of the first two, so only $x + y = 2$ and $x + z = 2$ remain. Writing $x = t$, the solutions are

$$(x, y, z) = (t,\ 2 - t,\ 2 - t), \qquad t \text{ any real number.}$$

The case $a = 1$

Now (1) and (2) are the same equation, $x + y + z = 2$, and the system is

$$x + y + z = 2, \qquad 2x + y + z = 2b.$$

Subtracting the first from the second gives $x = 2b - 2$, and then $y + z = 2 - x = 4 - 2b$. These two equations never contradict each other, so for *every* value of b there are infinitely many solutions:

$$(x, y, z) = (2b - 2,\ s,\ 4 - 2b - s), \qquad s \text{ any real number.}$$

Summary

- $a \neq 0, 1$: the unique solution $x = \dfrac{(a+1)b - 2}{a}$, $y = z = \dfrac{2-b}{a}$.

- $a = 0$, $b \neq 2$: no solution.
- $a = 0$, $b = 2$: $(t, 2 - t, 2 - t)$ for any t.
- $a = 1$, any b: $(2b - 2, s, 4 - 2b - s)$ for any s.

Checks

Take $a = 2$, $b = 4$: the formula gives $y = z = -1$ and $x = 5$, and indeed $5 - 1 - 2 = 2$, $5 - 2 - 1 = 2$ and $10 - 1 - 1 = 8 = 2b$. As a structural check, the general formula has a in the denominator, so it must break down at $a = 0$; it does, and our separate analysis shows why. At $a = 1$ the formula still makes sense and gives $(2b - 2, 2 - b, 2 - b)$, which is the member $s = 2 - b$ of our family: it is one solution among many, because $y = z$ is no longer forced. We verified all of the cases with exact rational arithmetic on many random values of a, b and the parameters.

Common slips

- Dividing by $1 - a$ or by a without saying that $a \neq 1$ or $a \neq 0$. Those divisions are exactly where the special cases hide.
- Assuming that a zero determinant means no solution. For $a = 1$ there are always infinitely many solutions, and for $a = 0$ there are some when $b = 2$.
- Forgetting to describe the infinite families explicitly: "infinitely many solutions" is not the same as solving the equations.

Beyond STEP

Each equation describes a plane in three dimensions, and the question asks how three planes can meet: in a single point, along a common line, or not at all (for $a = 0$, $b \neq 2$ the planes meet in pairs along three parallel lines, like the faces of a prism). The systematic tool is Gaussian elimination and the idea of the rank of a matrix: a system is consistent exactly when appending the right-hand side does not raise the rank. Strang's *Introduction to Linear Algebra* develops this picture beautifully.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.